

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DuCy _{HSE}	OSC_IN duty cycle	Digital mode (HSEYBYP = 1, HSEEXT = 1)	45	-	55	%
V _{HSE_ext_PP}	OSC_IN peak-to-peak amplitude	Analog mode (HSEYBYP = 1, HSEEXT = 0)	0.2	-	2/3 V _{DD}	V
V _{HSE_ext}	OSC_IN input range		0	-	V _{DD}	
t _r (HSE), t _f (HSE)	OSC_IN rise and fall time		0.05 / f _{ext_ext}	-	0.3 / f _{ext_ext}	

1. There is no specified rise and fall time for a digital input signal, but the V_{HSEH} and V_{HSEL} conditions must be fulfilled .
2. The DC component of the signal must ensure that the signal peaks are located between V_{DD} and V_{SS}.

Figure 14. AC timing diagram for high-speed external clock source (digital mode)

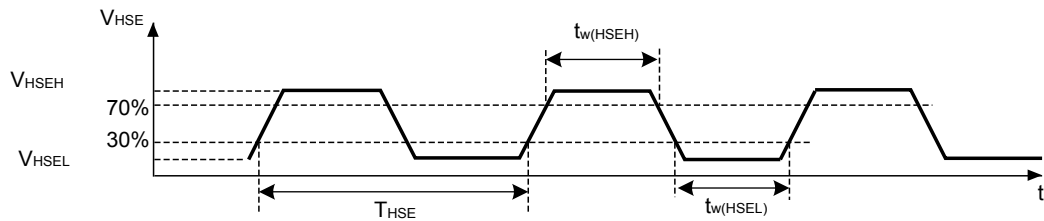


Figure 15. AC timing diagram for high-speed external clock source (analog mode)

